Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/16/2006
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Project Title:	Optical Characterization of Gas Phase, Gas Phase Chemistry, and Surface Chemistry during High Pressure Vapor Phase Deposition Processes under Laminar and Turbulent Flow Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bachmann, K. J. (North Carolina State University)		
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This research program established a novel growth technology, high-pressure chemical vapor deposition (HPCVD) for the growth of InN and related materials. HPCVD has been specifically developed to stabilize material systems with vast different vapor partial pressures as encountered in the growth of InN and related materials. The focus of this research program was to establish and demonstrate unique real-time optical monitoring tools that are capable to operate at elevated pressures and provide insights in the gas phase and surface chemistry during the growth process. The following research tasks have been accomplished: • The design, construction and testing of an advanced HPCVD reactor capable to operate at pressures up to 100 bar. • The design, construction and testing of a unique precursor injection system that allows to insert low-pressure organometallic precursors and gases in a high-pressure flow. • The development of principal-angle-reflectance spectroscopy (PARS) as a high-sensitive real-time optical monitoring tool, which allows the surface sensitive probing on of thin film nucleation and steady state growth at elevated pressures. • The design and

Task Description:	application of a pulsed-injection scheme that enabled the analysis of the reactor flow characteristics and the study of the decomposition kinetics of the precursors, ammonia and trimethylindium (TMI). • The highly surface sensitive probing on InN nucleation and steady state growth has been achieved by PAR, allowing the characterization of surface chemistry at a sub-monolayer level. • The growth and real-time monitoring of InN layers in the laminar flow regime, utilizing pressures up to 15 bars, flows from 2 slm to 8 slm, and a growth temperatures from 900K up to 1150K. • The structural and optical characterization of InN layers by XRD, Raman spectroscopy, Auger, absorption spectroscopy, spectroscopic ellipsometry, and reflectance spectroscopy. During the research program we demonstrated that HPCVD is a viable tool to optimize the process parameter for InN and indium rich In1-xGaxN alloys / heterostructures grown by HPCVD, utilizing an extended process window that is not accessible under MBE and MOCVD growth conditions. This research task target InN as model material, which, under conditions of conventional organometallic chemical vapor deposition (OMCVD / MOCVD) at sub-atmospheric pressure, is limited to processing temperatures around 800 K. We focus onto deposition temperatures > 900 K, which we consider as being essential for improving film quality and for improving the match between optimum deposition temperatures for heterostructural GaxIn1-xN/AlyGa1-yN layers. The desired range of processing temperatures may mandate operating pressures of up to 100 atm. In order to study and control the growth kinetics, novel real-time characterization techniques, together with a pulsed precursor injection scheme were developed. The optical access ports in the HPCVD reactor enable to use of several real-time optical techniques, capable to analyze the initial stages of heteroepitaxy and steady-state growth in the different pressure ranges. The demonstration and feasibility of these real-time diagnostics was the main objective of this research program. We not only showed that the optical diagnostic techniques provide viable insights in the growth process at elevated pressures, but also demonstrated that high-quality InN layers can be stabilized at reactor pressures around 15 bar and growth temperatures around 1150K. The significance of the research task is that it addresses the problems of optimization of flow conditions, containment of high pressure gases and real-time monitoring of deposition/etching kinetics under conditions not accessible with any currently applied growth method. High-pressure growth enables the control of stoichiometry of compounds at higher temperatures then accessible at subatmospheric pressure. This provides potential gains in perfection of the epilayers and their electrical and optical properties. It also will improve support of high density particles in fluidized bed processes for coatings, which pertain to improved fuel forms for NASA's interplanetary space exploration mission. The experiments addressing the real-time optical monitoring of indium nitride heteroepitaxy is expected to enhance the manufacturability for this important optoelectronic material in the context of advanced solid-state lasers, detectors and sensors. The work will provide insights into fundamental questions of heteroepitaxy related to the chemical and structural stability of heterointerfaces and mechanisms of defect formation and control of defect propagation. The integration of real-time monitoring techniques to characterize the gas phase kinetics and thin film formation process generates essential input parameter used in design and modeling of heterogeneous nucleation dynamics and thin film growth kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The unique HPCVD system developed during the research project enables the growth of high-quality InN layers at reactor pressures around 15 bar, and growth temperatures around 1150°C. These process conditions a close to those employed in GaN growth, enabling for the first time the growth of InN-GaN alloys and heterostructures, effectively suppressing the decomposition of indium-rich alloys at growth temperatures required for wide-bandgap group III-nitrides. The initial assessment showed that the structural quality is equivalent and even better then those values reported for MBE-grown InN material. Further improvements are expected by expanding the process parameter in the turbulent flow regime. The developed real-time optical monitoring techniques employed under HPCVD conditions exhibit the proposed superiority for optimizing and controlling the growth process. The developed tools are curial for the development of high-quality InN and related compound semiconductors as well as to gain insights in gas phase and surface chemistry processes during HPCVD. Device structures based on (Ga1-y-xAlyInx)N heterostructures and related alloys will enable the fabrication of • high-efficient energy conversion systems (multi-tandem solar cells & solid state lighting), • high speed optoelectronics for optical communication systems, • solid state lasers operating in the blue and ultraviolet regions, • spintronic and Terahertz device structures, and • nonlinear optical switching elements based on photonic band gap structures.
Task Progress:	During the research program, several real-time optical diagnostics have been development. Principal-angle-reflectance (PAR) spectroscopy has been introduced and successfully applied as a high surface sensitive probe, which enables to analyze the surface chemistry during InN nucleation and steady state growth at a sub-monolayer level. In addition, laser light scattering (LLS) was applied to characterize the surface morphology during nucleation and growth. The real-time optical monitoring techniques employed demonstrated their superiority in optimizing and controlling the growth process, as well as in gaining insight in gas phase and surface chemistry processes during HPCVD. The initial growth of InN by HPCVD has been assessed, demonstrating that high quality InN can be grown under high-pressure CVD conditions. The process parameters employed demonstrate that HPCVD is a viable tool for the growth indium rich In1-xGaxN alloys. We demonstrated that the high-pressure CVD approach enables the stabilization of highly volatile constituents as encountered in the growth of InN and related materials, extending the processing window to a region not accessible by MBE or low-pressure MOCVD.
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